Perimeter & Area in the Coordinate Plane — Practice — Key

A worked example comes before each type. Sketching the points on graph paper helps. Show your thinking.

Name: _____ Date: _____

Type 1 Horizontal & Vertical Lengths

Watch Me — Worked Example

Find the length from $(-2, 3)$ to $(5, 3)$.
Same y, so subtract the x's: $5 - (-2) = 7$. **Answer: 7**

1. $(1, 4)$ to $(9, 4)$

8

2. $(3, -2)$ to $(3, 6)$

8

3. $(-5, -1)$ to $(2, -1)$

7

4. $(-4, 7)$ to $(-4, -3)$

10

Type 2 Rectangles: Perimeter & Area

Watch Me — Worked Example

Rectangle $A(-1, 1)$, $B(5, 1)$, $C(5, 5)$, $D(-1, 5)$.
Width $= 5 - (-1) = 6$, height $= 5 - 1 = 4$. $P = 2(6 + 4) = 20$, $A = 6 \cdot 4 = 24$.
Answer: $P = 20$ units, $A = 24$ square units

5. $(0, 0)$, $(7, 0)$, $(7, 3)$, $(0, 3)$

$P = 20,\ A = 21$

6. $(-2, -1)$, $(4, -1)$, $(4, 4)$, $(-2, 4)$

$P = 22,\ A = 30$

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7. Square $(1,1)$, $(6,1)$, $(6,6)$, $(1,6)$

$P = 20,\ A = 25$

8. Rectangle $(0,0)$, $(9,0)$, $(9,k)$, $(0,k)$ has area 54. Find k $(k > 0)$.

$k = 6$

Type 3 Area of a Triangle

Watch Me — Worked Example

Triangle $(0,0)$, $(8,0)$, $(3,5)$.
The base lies on the x-axis: $b = 8$. The height is straight up to $(3,5)$: $h = 5$.
$A = \frac{1}{2} \cdot 8 \cdot 5$. **Answer: 20 square units**

9. $(1,1)$, $(7,1)$, $(4,6)$

15

10. $(-2,0)$, $(6,0)$, $(0,7)$

28

11. $(0,0)$, $(0,6)$, $(5,2)$ (hint: the base is vertical)

15

12. HARD $(2,-1)$, $(10,-1)$, $(12,4)$

20

Type 4 Slanted Sides: Use Pythagoras

Watch Me — Worked Example

Triangle $(0,0)$, $(3,0)$, $(0,4)$. Find the perimeter and area.
Legs 3 and 4; slanted side $\sqrt{3^2+4^2} = 5$. $P = 3+4+5 = 12$, $A = \frac{1}{2} \cdot 3 \cdot 4 = 6$.
Answer: $P = 12,\ A = 6$

13. $(0,0)$, $(5,0)$, $(0,12)$: find the perimeter.

$P = 30$

14. $(1,1)$, $(7,1)$, $(1,9)$: find the perimeter and area.

$P = 24,\ A = 24$

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15. $(-3, 0)$, $(3, 0)$, $(0, 4)$: find the perimeter and area.

$P = 16,\ A = 12$

16. HARD Square $(0, 0)$, $(3, 4)$, $(-1, 7)$, $(-4, 3)$: find the perimeter and area.

$P = 20,\ A = 25$

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